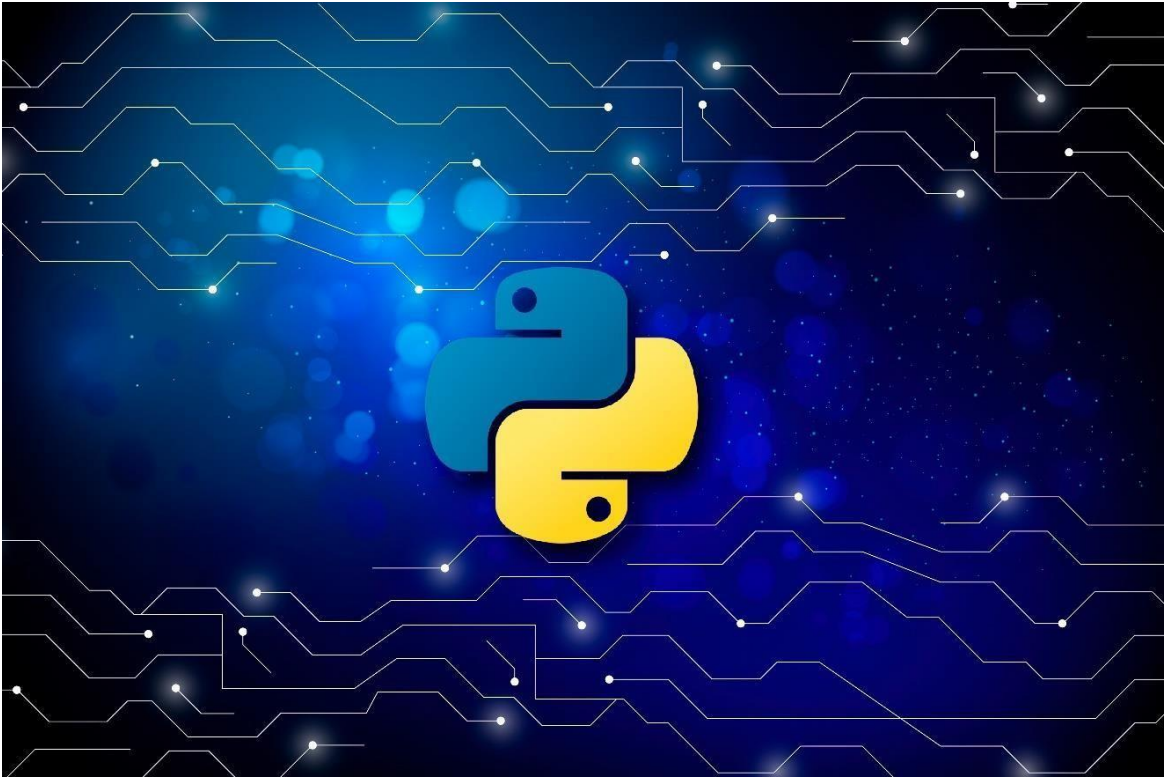




SSEEE1033 2024/2025-01

GROUP ASSIGNMENT

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Introduction:

Provided with an excel spreadsheet, we were required to analyze monthly climatological data for 3 cities spanning a 5year period from January 31,2018 to December 31,2022. The dataset captures two meteorological variables: temperature (°C) and precipitation(mm).

Following this data, we were presented with 4 objectives to perform in python.

- ❖ Task 1: Data Cleaning.: This process involves recognizing any abnormalities or extremities present in the data and essentially cleaning them by either removing them completely or using an appropriate imputation technique.
- ❖ Task 2: Descriptive Data Analysis: Here following the cleaned data we run the foundational statistics i.e. mean, median, standard deviation and correlation coefficients. This establishes the bulk of our data and presents judges any relationships between temperature and precipitation across the three cities.
- ❖ Task 3: Data Visualization: This step translates the statistics into intuitive graphical representations. Through time-temperature/precipitation graphs we identify seasonal cycles and any potential highlights are judged.
- ❖ Task 4: Predictive Analysis: Finally, we need to employ polynomial regression modeling to investigate and quantify relationships between temperature and precipitation. Furthermore, this model's performance is evaluated using metrics like Mean Squared Error, R-Squared to access its accuracy.

❖ Methodology:

Data preprocessing and cleaning served as the crucial first step before diving into the main analysis. Task 1 centered on inspecting the dataset for data quality issues, namely missing values and statistical outliers.

- First, the dataset was scanned for missing entries. Any gaps found were filled using the mean of their respective column to maintain data continuity. Next, the interquartile range (IQR) method was applied to detect outliers—values lying far outside the normal range. To prevent these extremes from distorting the analysis, they were replaced with the median of their column, a more robust measure against skewed distributions. [Ahmed Anwar Ali Sawi](#) led Task 1 and, as the first to contribute code, established the foundational data-cleaning framework that set the stage for all subsequent tasks. This established a cleaned datasheet usable by all future Tasks
- Task 2 was the bulk of data-processing. We performed two main types of analysis to understand the characteristics and relationships within our cleaned climate dataset. [Kareem Mohammed mahgoub Saeed](#) was in charge of this task. For each of the three cities, we calculated key statistical measures for both temperature and precipitation:
 1. Mean
 2. Median
 3. Standard Deviation

These metrics were computed individually for each variable and city, allowing us to compare climatic profiles and identify which locations exhibited greater variability or consistent patterns over time.

Afterwards we extracted and reported the correlation between each city's temperature and its corresponding precipitation. A value close to 1 suggested a strong correlation, a value close to -1 indicated a strong negative correlation while a value near to 0 suggested little to no linear correlation.

- Following the data analysis, in Task 3, [Musab Jamaleldien Mohamed Yousif](#) generated visualizations to better understand patterns, trends and differences in climate data across the 3 different cities. We generated a multi-panel plot with the top subplot displaying the trend of temperature in relation to time with a red line. The bottom subplot represented the trend of precipitation over time as a blue line. The top, bottom representation layout helps with easily recognizing patterns and behavior.

In accordance with the specifications we then designed a single line chart where each city's temperature was plotted with a distinct color. City-1 in red, City-2 in

blue and Ciy-3 in green. Together these represented the data processing achieved in task 2.

- Taking info from the visual graphs in Task 3, we're led to Task 4 where [MOSAB ABDELGADIR MAMOUN BAKER](#) had to model the relationship between temperature and precipitation, polynomial regression implementing Numpy. The temperature data from City 1 was selected as the predictor variable, with precipitation in the same city as the target variable. The appropriate polynomial degree was determined by visually inspecting the scatter plot of temp vs precipitation and testing different degrees. And finally the model's performance was evaluated using three different metrics:
 1. Mean Squared Error
 2. R-squared
 3. Mean Absolute Error
- And finally, the report was compiled and finalized by [Azwad Nur Naveed](#), who synthesized all team contributions into a unified and well-structured document. By consolidating the code, outputs, and analyses from each team member, I ensured both the codebase and final report were clear, consistent, and professionally presented. This process included thorough technical reviews to enhance code robustness, eliminate redundancies, and refine logic for clarity and correctness. Azwad also aligned the report with project objectives, performing meticulous checks for formatting, flow, and compliance with all submission requirements, delivering a polished and comprehensive final product.

❖ Code:

Here is the code we wrote for the assignment, separated into tasks due to easier readability and also limitations of the snipping software.

Task 1

Cleaning the datasheet and outputting a new cleaned datasheet named complete_data.

```
Task 1: Data Cleaning Process
-----
...

# reading the data from the csv file
curdir = os.path.dirname(__file__)
filepath = os.path.join(curdir, "Scenario 1 dataset.csv")
data = pd.read_csv(filepath, header=None, skiprows=1)
data = data.iloc[:, 1:]

#check for missing data
missing_values = data.isnull().sum()
is_missing = False
for i, val in enumerate(missing_values):
    if val != 0:
        print(f"Column {i+1} has {val} missing values")
        is_missing = True

complete_data = data.copy()
if is_missing:
    print("Filling missing values with column mean...")
    for col in complete_data.columns:
        # replace any instance of NaN by 'complete_data[col].mean()'
        complete_data[col] = complete_data[col].fillna(complete_data[col].mean())

#check for outliers and handle them
for col in complete_data.columns:
    Q1 = complete_data[col].quantile(0.25)
    Q3 = complete_data[col].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = (complete_data[col] < lower_bound) | (complete_data[col] > upper_bound)
    if outliers.sum() != 0:
        print(f"Column {col}: {outliers.sum()} outliers detected")
        print("Replacing Outliers with Column median.....")

    # Replace outliers with column median
    median = complete_data[col].median()
    complete_data.loc[outliers, col] = median
```

Task 2

Bulk of statistical operations, finding mean, median and standard deviation based on values from 'complete_data' and also calculating coefficients.

```
59 Task 2: Descriptive Data Analysis
60 -----
61
62 ...
63 #performing the summary statistics
64 c1t_mean = complete_data[1].mean()
65 c1t_median = complete_data[1].median()
66 c1t_std = complete_data[1].std()
67
68 c2t_mean = complete_data[2].mean()
69 c2t_median = complete_data[2].median()
70 c2t_std = complete_data[2].std()
71
72 c3t_mean = complete_data[3].mean()
73 c3t_median = complete_data[3].median()
74 c3t_std = complete_data[3].std()
75
76
77
78 c1p_mean = complete_data[4].mean()
79 c1p_median = complete_data[4].median()
80 c1p_std = complete_data[4].std()
81
82 c2p_mean = complete_data[5].mean()
83 c2p_median = complete_data[5].median()
84 c2p_std = complete_data[5].std()
85
86 c3p_mean = complete_data[6].mean()
87 c3p_median = complete_data[6].median()
88 c3p_std = complete_data[6].std()
89
90 print(f'1st city temperatures: mean = {c1t_mean}, median = {c1t_median}, standard deviation = {c1t_std}, precipitation: mean = {c1p_mean}, median = {c1p_median}, standard deviation = {c1p_std}')
91 print(f'2nd city temperatures: mean = {c2t_mean}, median = {c2t_median}, standard deviation = {c2t_std}, precipitation: mean = {c2p_mean}, median = {c2p_median}, standard deviation = {c2p_std}')
92 print(f'3rd city temperatures: mean = {c3t_mean}, median = {c3t_median}, standard deviation = {c3t_std}, precipitation: mean = {c3p_mean}, median = {c3p_median}, standard deviation = {c3p_std}')
93
94 #calculating the correlation coeff
95
96 correlation_matrix = complete_data.corr(method="pearson")
97
98 city1_corr = correlation_matrix.iloc[0,3]
99 print(f'1st city correlation coefficient = {city1_corr}')
100
101 city2_corr = correlation_matrix.iloc[1,4]
102 print(f'2nd city correlation coefficient = {city2_corr}')
103
104 city3_corr = correlation_matrix.iloc[2,5]
105 print(f'3rd city correlation coefficient = {city3_corr}')
```

Task 3

Code for presenting the graphical representation of the values from task2. Dividing the data into subplots for temperature and precipitation and one large line graph for comparison between all 3 cities.

```
109 -----
110 TASK 3 : data representation
111 -----
112 '''
113 import matplotlib.pyplot as plt
114 cities= ['city1','city2','city3']
115
116 for i in range(3):
117     plt.subplot(6,1,2*i+1)
118     #teperature plots
119
120     plt.plot(complete_data[i+1],'-r',label='temperature')
121     plt.ylabel('Temperature(°c)')
122     plt.title(f'temp for city {cities[i]}')
123     plt.legend()
124     plt.grid(True)
125
126     #precipiation plot
127
128     plt.subplot(6,1,2*i+2)
129     plt.plot(complete_data[i+4],'-b',label='precipitation')
130     plt.title(f'{cities[i]} precipitation')
131     plt.grid(True)
132
133 plt.tight_layout()
134 plt.show()
135
136 '''
137
138 TASK 3: comarative temp plot
139
140 '''
141 plt.plot(complete_data[1],'-r',label='city1')
142 plt.plot(complete_data[2],'-b',label='city2')
143 plt.plot(complete_data[3],'-g',label='city3')
144 plt.title('comarative temp plot across cities')
145 plt.xlabel('days')
146 plt.ylabel('temperature')
147 plt.legend()
148 plt.grid(True)
149 plt.show()
150
```

Task 4

This code builds and evaluates a polynomial regression model to predict precipitation in City 1 using its temperature as the predictor. We needed to implement the Numpy Lib. It fits a second-degree polynomial, calculates error metrics (MSE, MAE, R^2), and visualizes the relationship between temperature and precipitation.

```
TASK 4: Prediction
-----
Polynomial regression using temperature to predict precipitation
...

# Predictor (x): temperature of city 1
x = complete_data[1].values

# Target (y): precipitation of city 1
y = complete_data[4].values

degree = 2 # you can try 1, 2, 3 and choose best based on metrics

coeffs = np.polyfit(x, y, degree) # fit polynomial
model = np.poly1d(coeffs)         # build polynomial function

y_pred = model(x)

# Mean Squared Error (MSE)
mse = np.mean((y - y_pred)**2)

# Mean Absolute Error (MAE)
mae = np.mean(np.abs(y - y_pred))

# R-squared ( $R^2$ )
ss_res = np.sum((y - y_pred)**2)
ss_tot = np.sum((y - np.mean(y))**2)
r2 = 1 - (ss_res / ss_tot)

print("\n--- Task 4: Polynomial Regression (City 1) ---")
print("Polynomial degree:", degree)
print("Model coefficients:", coeffs)
print("MSE:", mse)
print("MAE:", mae)
print("R-squared:", r2)

import matplotlib.pyplot as plt

# Sort x for a smooth curve
idx = np.argsort(x)
x_sorted = x[idx]
y_sorted = y[idx]
y_fit_sorted = model(x_sorted)

plt.scatter(x, y, label="Actual data")
plt.plot(x_sorted, y_fit_sorted, label=f"Poly fit (deg={degree})")
plt.title("City 1: Precipitation vs Temperature (Polynomial Regression)")
plt.xlabel("Temperature (°C)")
plt.ylabel("Precipitation")
plt.grid(True)
plt.legend()
plt.show()
```


➤ Graphs and Visualizations

Running the code, we get graphical representation of our data values as an output from the complete_datasheet.

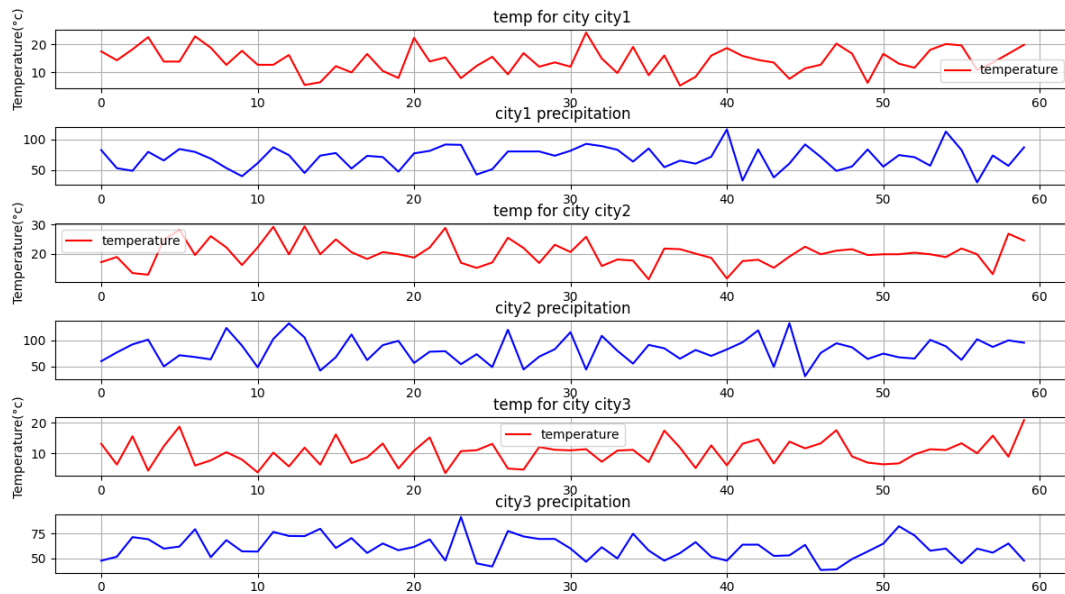


Fig: Visual Data of Temperature and Precipitation over time for all 3 cities.

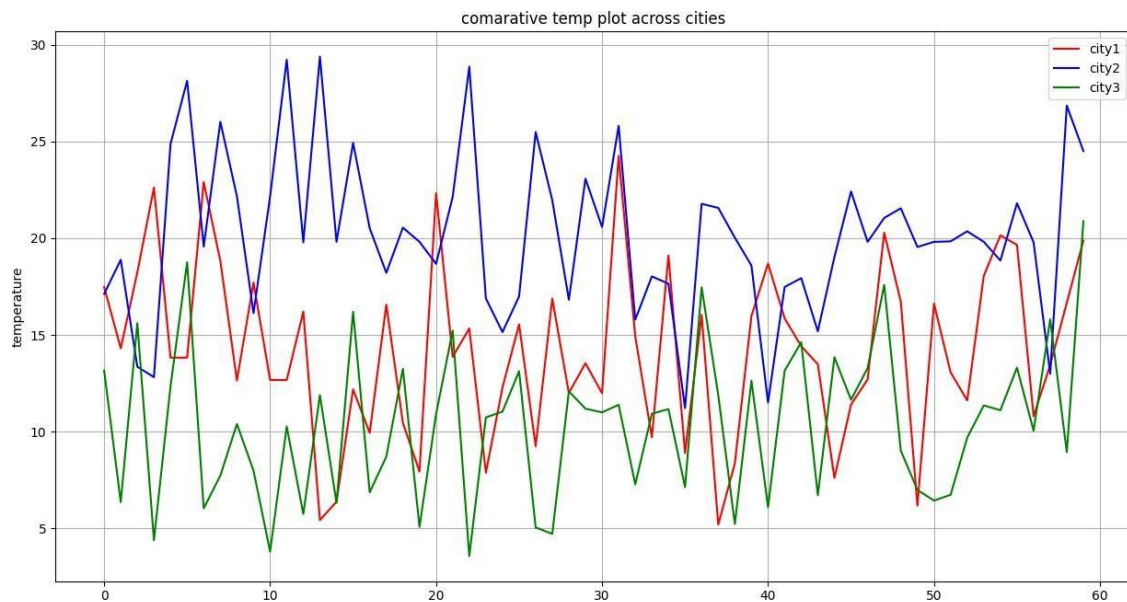


Fig: Comparative plot of temperature/precipitation of all 3 cities.

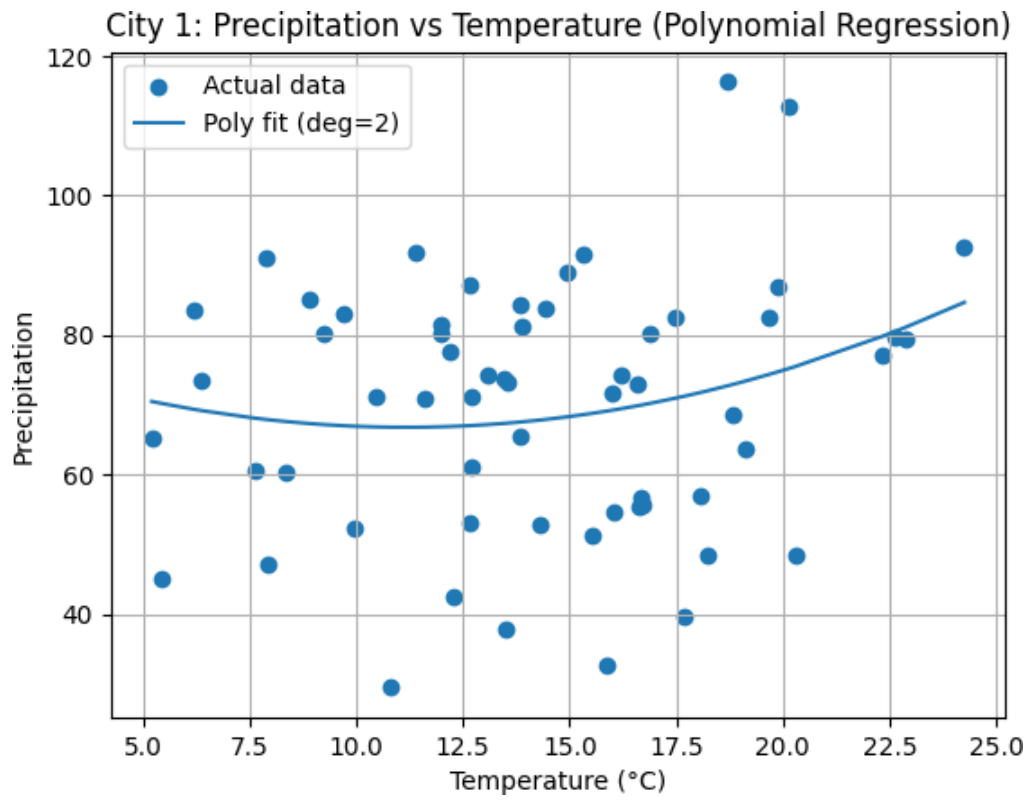


Fig: Polynomial Regression for City 1 based on the predictive model.

❖ Discussion and Explanations:

Separating our results into points :

- The data cleaning process revealed underlying quality issues requiring intervention to ensure the reliability of subsequent analyses. The detection of five outliers in City 2's temperature data suggests this city may experience more extreme or variable thermal conditions compared to the others. This was addressed by imputing the median value, thereby preserving the central tendency of the data while minimizing distortion.

Additionally, single outliers were found in the precipitation records of City 1 and City 2. These could represent legitimate extreme weather events (e.g., intense rainstorms) or measurement errors. By replacing them with the median, we effectively "smoothed" the dataset for general trend analysis, though future work could investigate whether these outliers correspond to documented meteorological events.

- City 2 emerges as the warmest and wettest location, with a mean temperature of 20.18°C and mean precipitation of 80.89 mm. However, its precipitation also shows the highest variability (standard deviation of 23.87 mm), indicating an unstable or seasonally concentrated rainfall pattern, potentially characteristic of a monsoonal or convective climate.

City 1 displays a temperate profile, with moderate mean temperature (14.23°C) and precipitation (69.91 mm). Its temperature standard deviation (4.54°C) is the highest among the three, suggesting greater daily or seasonal thermal fluctuations. City 3 is the coolest and driest, with the lowest mean temperature (10.37°C) and precipitation (60.57 mm). Notably, it also shows the lowest precipitation variability (standard deviation of 11.42 mm), implying a more consistent, possibly maritime or higher-latitude climate with steadier rainfall distribution.

- The relationship between temperature and precipitation in City 1 was modeled using a second-degree polynomial regression of the form:

$$\text{Precipitation} = 0.1043 \times (\text{Temp})^2 - 2.3268 \times (\text{Temp}) + 79.7884$$

This quadratic model suggests a curvilinear relationship where precipitation initially decreases with increasing temperature, reaches a minimum, and then increases slightly. The positive quadratic coefficient (0.1043) indicates an upward curvature at higher temperatures.

Further Predictive Analysis using the model and using different metrics, Mean Squared Error: 309.74 is a relatively high value that represents a substantial prediction error.

Mean Absolute Error: 14.54 On average, model predictions deviate from actual precipitation values by approximately 14.54 mm.

R-Squared: 0.047 The model explains only 4.7% of the variance in precipitation, indicating very limited predictive power.

➤ Conclusion:

To conclude our findings, City 2 emerged as the warmest of the 3 while City 3 was the driest and the coolest with the most stable rainfall patterns.

The lack of correlation from the analysis indicates that temperature alone is insufficient as a predictor for precipitation. This is directly supported from the minimal explanatory power the model based on City 1 achieved.

This leads us to believe that local precipitation is governed more by complex multi-factor processes rather than simply temperature. Using the methodology of this code and the analytical value can be used as a foundation for further more multi-factor and complex modeling.

References:

- Video link

https://drive.google.com/file/d/1N92YInoTXheWG6QNy_s9MHBRcsVOSq3dl/view?usp=sharing

- Subject lectures.
- Wilks, D. S. (2011). Statistical Methods in the Atmospheric Sciences (3rd ed.)
- Tomczak, M., & Godfrey, J. S. (2003).